YIFEI SUN

Staff Full-Stack Engineer · Backend, Frontend & Cloud Infrastructure · 13 Years

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SUMMARY

Full-stack engineer with 13 years building customer-facing applications and cloud-native backend services across the entire stack. Deep experience in backend systems (Java, Node.js, REST APIs, event-driven architecture) paired with extensive frontend delivery (Angular, TypeScript, JavaScript) across customer applications, admin dashboards, and mobile platforms. Led cross-team redesign of a high-traffic API serving millions of daily requests (p99 250ms→50ms, availability 98%→99.9%). Combines hands-on full-stack engineering with cloud infrastructure expertise (AWS, Kubernetes, Terraform) and cross-organizational technical leadership.

LEADERSHIP HIGHLIGHTS

- Led cross-team API redesign: p99 5× reduction, 99.9% availability, Kubernetes-orchestrated microservices on AWS
- Built cloud infrastructure from scratch — AWS provisioning (Ansible/Terraform), SQS/SNS event pipelines, API gateways, service mesh
- Full-stack across every role: customer-facing apps (Angular, TypeScript, Android), admin dashboards, Node.js BFF, backend services
- Mentored senior engineers to Principal-level scope; drove architecture standards and observability practices across 10+ teams

Technologies: Backend: Java, Node.js, Python, Spring Boot · **Frontend:** Angular, TypeScript, JavaScript, jQuery, Android · REST API, GraphQL (Netflix DGS), gRPC · **AWS** (EC2, ECS, EKS, SQS, SNS, S3, RDS, DynamoDB), Terraform, Ansible · Kubernetes, Docker, Kafka, Redis, PostgreSQL, Cassandra · Microservices, Event-Driven, CQRS · Datadog, Grafana · CI/CD

EXPERIENCE

CarGurus — Principal Software Engineer / Technical Lead

Oct 2019 – Present

High-Traffic API Modernization — Full-Stack Cloud Services

- Led 3 principal/senior engineers and partnered with ~10 teams across 3 departments to redesign the core API behind CarGurus' most-visited page (~8,000 rpm, millions of daily impressions), deploying on AWS with Kubernetes-orchestrated microservices.
- Introduced a new orchestration service consolidating 12 downstream integrations into 5 well-defined services with circuit breaking, fault isolation, and idempotent retry.
- Built automated parity-testing infrastructure comparing new vs. legacy responses at 50% production traffic, enabling safe, data-driven rollout decisions.

p99: 250 ms → 50 ms (5×) **Availability: 98% → 99.9%** **Deploy: 1 hr → 3 min** (20×)

Dealer-Facing Admin Dashboard & GraphQL Federation

- Built dealer-facing admin dashboards for inventory and listing management, serving as the primary interface for CarGurus' B2B customers.
- Designed and integrated a Netflix DGS subgraph into the federated GraphQL gateway, exposing vehicle delivery data to frontend, backend, and AI consumers from a single graph.

Digital Commerce Platform — Event-Driven Architecture

- Led backend design for a new commerce product on AWS using microservice and event-driven architecture (SQS/SNS, Kafka) with Terraform-managed infrastructure.
- Built infrastructure for rapid product experimentation — enabling quick launch, measurement, and safe retirement of product bets without destabilizing core systems.

Cross-Org Platform Leadership & Standards

- Established patterns for API design, observability, caching, and distributed orchestration adopted across multiple teams; authored design docs, RFCs, and decision records.
- Led 10+ training sessions for engineering teams. Mentored senior engineers on system design and path to Principal-level scope.

Applause — Senior Software Engineer / Technical Lead

Oct 2017 – Oct 2019

Full-Stack Customer Applications & Authentication Modernization

- Built customer-facing applications and admin dashboards end-to-end using Angular, TypeScript, and Node.js BFF (backend-for-frontend) across the full stack.
- Upgraded legacy authentication to a modern Keycloak-based SSO platform, with custom token authentication plugins for Kong API gateway written in Python. Unified auth across internal systems, enterprise partners, and external community.
- Set up API gateway and service mesh infrastructure (Kong + Linkerd on Kubernetes) with Glassfish + JVM performance tuning.

Event Sourcing & Cross-Org Collaboration

- Drove design and prototyping of microservice architecture based on event sourcing and CQRS; defined foundational patterns adopted by new domains.
- Partnered with product, support, and customer-facing teams to modernize internal workflows and enterprise integrations.

Toast Inc. — Senior Software Engineer

Sep 2014 – Oct 2017

Full-Stack Product Development — Restaurant Platform

- Among the earliest engineers during hypergrowth; built core services end-to-end (Order Service, Labor Service, REST APIs) and customer-facing admin dashboards (jQuery, Angular) for restaurant owners.
- Developed consumer-facing Android application for Toast's mobile ordering platform.
- Built a custom API gateway in Nginx with Lua scripting for rate limiting, request routing, and traffic management.
- Set up the initial AWS environment using Ansible — provisioned infrastructure, deployment pipelines, and monitoring before a dedicated DevOps team existed.

Third-Party Integrations & Cross-Functional Leadership

- Led integrations with key partners (e.g., workforce management), ensuring reliable bidirectional sync. Partnered with PMs, sales, and support to translate early customer feedback into platform capabilities.

The MathWorks Inc. — Software Engineer II

Jul 2013 – Aug 2014

Developed internal engineering tools (Java backend) for MATLAB/Simulink workflows. Improved backend automation systems for build/test reliability and developer productivity.

EDUCATION

M.S. in Electrical and Computer Engineering — Northeastern University, Boston 2010 – 2013

B.S. in Information Science — Beijing University of Aeronautics and Astronautics 2006 – 2010